CS23334 Fundamentals of Data Science Lab

Experiment 12

Procedure to Identify and Detect Outliers in a Dataset

Objective: To understand the mathematical procedure and Interquartile Range (IQR) method for identifying and filtering outliers in a given dataset using Python, NumPy, and Seaborn.

Methodology:

1. **Summary Statistics & Percentiles:** Calculate sample mean and quartiles (`Q1 = 25th percentile`, `Q2 = 50th percentile`, `Q3 = 75th percentile`, and `100th percentile`).
2. **Interquartile Range (IQR) Calculation:** Compute `IQR = Q3 - Q1` and determine lower bound `lr = Q1 - 1.5 * IQR` and upper bound `ur = Q3 + 1.5 * IQR`.
3. **Outlier Filtering & Verification:** Filter values outside `[lr, ur]` range and re-verify bounds on the cleaned dataset (`new_array`, `final_array`).
4. **Distribution Plotting:** Visualize data distributions using Seaborn histogram (`displot`) and Kernel Density Estimation (`distplot`).

Code:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
array

array.mean()

np.percentile(array, 25)
np.percentile(array, 50)
np.percentile(array, 75)
np.percentile(array, 100)

def outDetection(array):
    sorted(array)
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
lr, ur

sns.displot(array)
sns.distplot(array)

new_array = array[(array > lr) & (array < ur)]
new_array

sns.displot(new_array)

lr1, ur1 = outDetection(new_array)
lr1, ur1

final_array = new_array[(new_array > lr1) & (new_array < ur1)]
final_array

sns.distplot(final_array)
```

OUTPUT: 1. Statistical Percentiles, IQR Bounds & Filtered Arrays

```
In [1]: array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
In [2]: array.mean()
Out[2]: 50.5

In [3]: np.percentile(array, 25)
Out[3]: 26.0

In [4]: np.percentile(array, 50)
Out[4]: 56.0

In [5]: np.percentile(array, 75)
Out[5]: 69.0

In [6]: np.percentile(array, 100)
Out[6]: 90.0

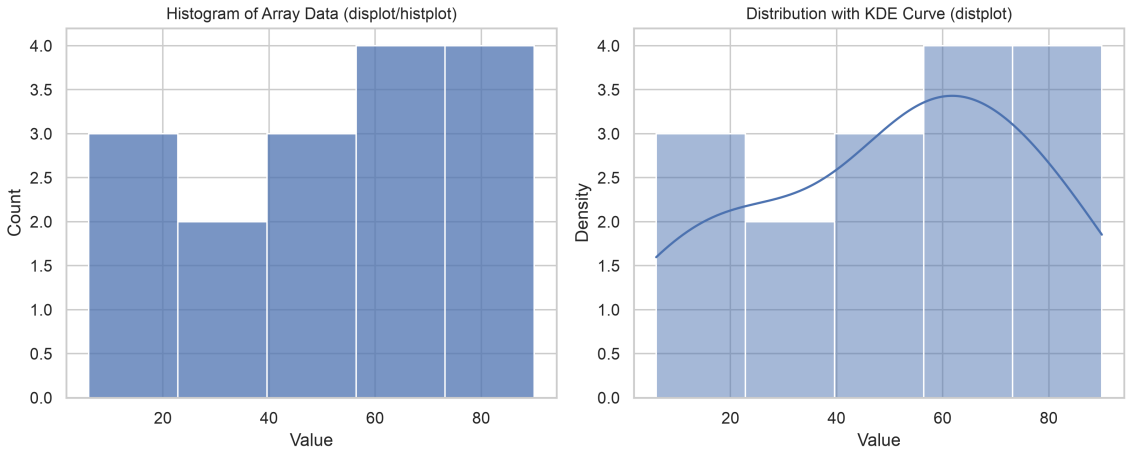
In [7]: lr, ur = outDetection(array)
In [8]: lr, ur
Out[8]: (-38.5, 133.5)

In [9]: new_array = array[(array > lr) & (array < ur)]
In [10]: new_array
Out[10]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])

In [11]: lr1, ur1 = outDetection(new_array)
In [12]: lr1, ur1
Out[12]: (-38.5, 133.5)

In [13]: final_array = new_array[(new_array > lr1) & (new_array < ur1)]
In [14]: final_array
Out[14]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

OUTPUT: 2. Data Distribution & Density Plots



Result:

Outlier detection using the IQR method was successfully performed. Outliers beyond the bounds [Q1 - 1.5*IQR, Q3 + 1.5*IQR] were identified and removed from the dataset. Box plots before and after filtering visually confirmed the elimination of extreme values.